

Yosief Abraham

Dr. Ashby Monk

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Understanding the Total Portfolio Approach in Institutional Investing

Overview

The purpose of this research is to examine the Total Portfolio Approach (TPA) as an alternative framework for institutional portfolio construction and governance. While TPA has gained prominence among a small group of large sovereign wealth funds and public pensions, its adoption remains limited and uneven. This paper analyzes how and why TPA has emerged, the conditions under which it has been successfully implemented, and the institutional capabilities required to support it.

This paper will contrast TPA with traditional Strategic Asset Allocation (SAA) and evaluate its real-world application through case studies of the Australia's Future Fund, Canada's CPP Investments, and Singapore's GIC; focusing on governance, risk management, and organizational design. It then examines California's CalPERS planned transition toward TPA, using it to highlight both the opportunities and constraints of adopting TPA in politically visible institutions.

Total Portfolio Approach (TPA)

A Total Portfolio Approach, or TPA, is a dynamic framework for managing investments holistically, rather than through static asset class silos. In a TPA model, every investment decision is evaluated based on its impact on the portfolio's overall goals and risk profile (Elkamhi). The most widely adopted investment structure, Single Asset Allocation, or SAA, takes a different approach by adhering to fixed percentage allocations for each asset class.

With a TPA, the chief investment officer, or CIO, would have the flexibility and autonomy to allocate capital across assets, guided by how each investment contributes to the entire fund's objectives (Davidow). This approach often emphasizes a factor-based perspective by looking at underlying risk factors like growth and inflation, while encouraging investment opportunities across different assets to compete based on merit in improving the total portfolio. In short, TPA is goal-driven and empowers the CIO: success is measured by the portfolio's total return relative to its goals, and the investment process is more nimble and integrated across various asset classes (Hodgson).

The current industry recognizes TPA as an "evolution" beyond traditional modern portfolio theory practices. This evolution is guided by insights of rigid asset allocations limiting portfolio responsiveness and overall performance. To dive deeper, research led by Modern Portfolio Theory (MPT) historically supported fixed strategic allocations, but TPA seeks to overcome some of MPT's practical limitations (Elkamhi). In the modern world, real-time data and advanced risk systems enable TPA's to conduct dynamic decision-making. Instead of focusing solely on beating asset-class benchmarks, TPA focuses on achieving the fund's specific goals regarding meeting liabilities or a target return (Resonanz Capital). Importantly, TPA also

requires cultural and governance shifts, where boards must delegate more discretion to investment professionals. Beyond increased autonomy of investment professionals, teams spanning different asset classes must collaborate as “one portfolio” rather than protect individual asset class mandates (Hodgson). Overall, the TPA is understood as a flexible, whole-portfolio mindset that leverages technology, broad asset classes, and active risk management to adapt to changing market conditions and opportunities.

Applications of TPA

TPA’s flexibility makes it particularly useful in complex, rapidly changing investment environments (Davidow). Key applications of the Total Portfolio Approach include dynamic asset allocation, holistic evaluation of investments, and goal focused investing.

Dynamic Asset Allocation: Because TPA isn’t locked into a fixed preset asset mix, investment teams are able to adjust allocations in real time to best respond to market conditions and risks (Two Sigma Venn). For example, under TPA a fund can quickly reduce bond exposure when interest rates spike or increase commodities exposure when inflation risks rise. In a traditional portfolio (SAA), investors are unable to temporarily deviate from the fund-driven asset allocations, resulting in limited defensive strategies to various risks and capped returns (Hodgson). This agility is valuable for navigating market volatility, regime shifts, or crises.

Holistic Evaluation of Investments: In TPA, investments are chosen based on their contribution to the portfolio returns, regardless of their asset category. Because of this, investments can be evaluated by its factor exposures regarding equity risk, credit risk, illiquidity

premium, and how it diversifies or enhances the total portfolio (Davidow). This helps funds hold higher allocations to alternative investments and manage the entire portfolio risks without the constraint of a strict asset allocation split

Goal Driven Investing: Many pension plans and endowments use TPA to focus on final outcomes regarding spending targets and meeting liabilities. By having the larger focus fixated on overall performance, various hedging strategies are utilized: examples of unique investment/hedging strategies can include the use of derivatives (Two Sigma Venn). In the end, investors are enabled to maximize the probability by actively steering the portfolio as conditions evolve, rather than following a static course.

Strategic Asset Allocation (SAA)

Strategic Asset Allocation (SAA) is the traditional approach to portfolio construction where a fund defines a fixed, long-term target mix of asset classes, with infrequent periodic rebalancing. For example, a classic SAA might dictate 60% stocks, 30% bonds, 10% alternative assets, and the portfolio is managed to maintain those weights over time as needed. This structure is often enshrined as an investment policy, serving as the portfolio's "North Star" or benchmark. Performance success in an SAA framework is usually judged relative to these benchmarks; each asset class often has its own benchmark (e.g. an equity index for stocks) and managers aim to outperform within their sleeve. At the total fund level, the main goal is to achieve the overall return target by adhering to the asset mix and rebalancing periodically (Van Inwegen).

SAA is typically board-driven: a governing board or investment committee sets the asset allocation policy and reviews it only occasionally: annually or every few years. By design, SAA deliberately avoids frequent changes, under the premise that market timing and constant reallocation can harm long-run results. Instead, SAA relies on the idea that a balanced, diversified mix will ride out short-term volatility. For instance, even if markets become turbulent, an SAA-guided fund usually rebalances back to its target weights rather than dramatically shifting stance. This rigidity brings simplicity and clarity: it's easy for stakeholders to understand and to hold managers accountable to specific benchmarks. Historically, SAA has been considered the “gold standard” of institutional investing governance (Barsoum). It fits the governance model of periodic board meetings and limited technology, where historically speaking, dynamic and real-time adjustments were not feasible for many funds. SAA's focus on long-term averages and stable policy targets made sense when data and systems to support rapid shifts were lacking. Even today, SAA remains the predominant approach: by some estimates, over 95% assets under management use a strategic asset allocation model (Jarvis).

Applications of SSA

Strategic Asset Allocation is utilized in a wide range of investment contexts, especially where stability, governance oversight, and simplicity are paramount. Key applications and contexts include pensions funds, insurance portfolios, endowment/foundations, individual investors, and sovereign funds (Van Inwegen).

Pension Funds and Insurance Portfolios: Traditional pension funds typically employ SAA to balance growth assets and liability-matching assets. This structure is aligned with the fund's

liability profile, where the long-term policy helps the fund demonstrate prudence and consistency to regulators and stakeholders.

Endowments and Foundations: Many endowments use SAA as a way to maintain spending objectives in perpetuity. An endowment might adopt, say, 70% growth assets (equities, private equity, etc.) and 30% defensive (bonds, cash) as a strategic mix to target a long-run ~5% real return to support annual spending. The discipline of SAA ensures the endowment doesn't take on too much or too little risk relative to its mandate.

Individual Investors and 60/40 Portfolios: The classic 60/40 portfolio of 60% stocks and 40% bonds, is essentially an SAA concept applied to individuals and balanced mutual funds. Many financial advisors set a strategic mix for clients based on risk tolerance and update it only as clients' life stages or goals change (*Morgan Stanley Research*).

Sovereign Wealth and Reserve Funds: Some sovereign wealth funds and central bank reserves use SAA-like approaches. They define, for example, a fixed split between domestic and international assets, or between equities and fixed income, to reflect national objectives and risk tolerance. By keeping a steady allocation, they avoid political pressures to time markets (Gorshkov).

In summary, SAA is applied when consistency, clarity, and ease of governance are prioritized. It works best in relatively stable environments or when the primary risks are well-understood and do not require frequent re-optimization.

TPA vs. SAA

Both TPA and SAA are high-level strategies for portfolio construction, but they differ fundamentally in philosophy and execution. In essence, SAA is a static blueprint while TPA is a flexible, goal-focused process. The table below summarizes key differences between the two approaches:

	Strategic Asset Allocation (SAA)	Total Portfolio Approach (TPA)
Primary Objective	Assessed relative to a benchmark portfolio or policy index.	Judged by total fund outcomes against those goals.
Success Measured By	Relative returns vs. benchmarks	Absolute total portfolio return and ability to meet objectives
Investment Silos	Organized by asset class	Organized by risk factors/strategies
Allocation Decisions	Board-driven.	CIO and team-driven

Portfolio Implementation	Multiple teams	One unified team
Rebalancing & Adjustments	Static with periodic rebalancing	Dynamic and continuous without the need to adhere to preset weights.

Australia Future Fund

The Future Fund is Australia’s sovereign wealth fund. It was established in 2006 under the Future Fund Act 2006 to strengthen Australia’s long-term financial position and address the government’s unfunded pension liabilities (*Future Fund Australia Sovereign Wealth Fund*). The Fund was created during a period of strong fiscal capacity, when Australia had the opportunity to convert their budget surpluses and asset-sale proceeds into a permanent national balance-sheet asset. The Fund’s mandate is to preserve and enhance the real value of its capital over the long term, with a target return of CPI + 4% to 5% per annum (or long-term target of ~6.9% per annum), while taking an “acceptable but not excessive” level of risk (*Future Fund Australia Sovereign Wealth Fund*). The Future Fund is not designed as a short-term stabilization tool; instead, it is explicitly intended to compound over decades and support future fiscal obligations. As of June 2025, the Future Fund managed approximately A\$252 billion (USD \$167.6 billion), making it the Australian Government’s largest financial asset (*Future Fund Australia Sovereign Wealth Fund*).

Since inception in 2006, the Fund has exceeded the mandated CPI + 4–5% target, delivering annualized returns comfortably above inflation while maintaining relatively low volatility for a growth-oriented portfolio (*Future Fund Australia Sovereign Wealth Fund*).

The Future Fund maintains a highly diversified, global portfolio spanning public and private markets. Unlike traditional pension funds built around static strategic asset allocations, the Future Fund applies a Total Portfolio Approach, where asset classes compete for capital based on their contribution to overall portfolio objectives rather than predefined weights (*Future Fund Australia Sovereign Wealth Fund*). As of June 2025, the portfolio consisted broadly of global equities, private equity, infrastructure, timberland, real estate, credit strategies, alternatives, and cash for liquidity and optionality (*Future Fund Australia Sovereign Wealth Fund*).

Equities form the largest return driver in the portfolio, with a relatively smaller allocation to traditional government bonds. This reflects its view that bonds may not provide sufficient diversification or inflation protection in certain macro regimes. Instead, diversification is met through a wider toolkit that includes real assets, alternatives, currency exposures, and dynamic positioning. This structure allows the Fund to adapt as correlations shift rather than relying on a typical fixed 60/40-style framework.

Geographically, the portfolio is overwhelmingly international with limited home bias toward Australia. The rationale is structural: Australia represents a small share of global opportunity sets, and concentrating domestically would introduce unnecessary macro and fiscal correlation risk. While the Fund does invest in Australian infrastructure and energy-related assets, the majority of capital is deployed across North America, Europe, Asia, among other

markets (*Future Fund Australia Sovereign Wealth Fund*). Additionally, currency exposure is actively managed relative to the Australian dollar, balancing diversification benefits with the Fund's domestic purchasing-power objective.

In recent years, the Future Fund has benefited from strong global equity performance, particularly following the post-pandemic recovery. While short-term returns have fluctuated with markets, long-run results remain the central measure of success. Over rolling 10 and 15 year periods, the Fund has delivered real returns that exceed its mandate benchmark of 6.9% per annum, while maintaining drawdown characteristics consistent with its risk appetite (*Future Fund Australia Sovereign Wealth Fund*). Here is more context on the fund returns:

- FY2025 (to 30 June 2025): Future Fund delivered a 12.2% annual return. (Duke).
- 10-year annualized return (to 30 June 2025): approximately 8.0% per annum (Duke).
- 10-year average (to 30 June 2024): 8.3% per annum (Duke).

Despite this strong performance, the Board is cautious of forward-looking expected returns becoming lower and more volatile due to elevated valuations, higher interest rates, and geopolitical uncertainty. Risk management here focuses on assessing how exposures interact under different economic regimes. This approach allows strategies that may underperform in isolation to be retained if they meaningfully reduce portfolio drawdown risk or improve resilience in stressed environments.

The Future Fund uses a reference portfolio concept and risk tolerance framework to anchor governance while preserving flexibility (*Future Fund Australia Sovereign Wealth Fund*). The reference portfolio is not a benchmark to be tracked, but a tool to express the Fund's long-term risk posture and to help the Board assess whether the actual portfolio is taking more or

less risk than intended. Active risk is managed by the investment team reallocating capital dynamically as conditions change. Furthermore, capital preservation is not defined as avoiding volatility. As a result, the Fund is willing to underperform in certain bull markets to reduce vulnerability in adverse scenarios.

Since its inception, the Future Fund has evolved substantially in both scale and sophistication. Initially, the Fund's task was to deploy its A\$60.5 billion (roughly USD \$46 billion) seed capital into global markets. Over time, it expanded meaningfully into private equity, infrastructure, and alternatives as internal capabilities and governance matured. A key evolution occurred as the Fund embraced a total portfolio mindset, moving away from rigid asset-class silos toward integrated risk management and dynamic capital allocation (Hutchens). Another major inflection point came when the government deferred withdrawals until at least 2032–33, which allowed for the Fund to remain in compounding mode for longer, materially increasing its projected long-term growth (*Future Fund Australia Sovereign Wealth Fund*).

The Future Fund is governed by the Future Fund Board of Guardians, an independent board composed of experienced investment and financial professionals. Day-to-day operations are executed by the Future Fund Management Agency, or FFMA. The organization employs roughly 300–350 staff, making it very lean given the scale of assets under management (*Future Fund Australia Sovereign Wealth Fund*). Most employees are investment-facing, supported by risk, legal, operations, and technology functions.

The Fund is headquartered in Australia, with offices in Melbourne and Sydney, and does not operate permanent overseas offices. Instead, it achieves global reach through extensive engagement with external managers and partners, supplemented by frequent international travel

for due diligence and relationship management (*Future Fund Australia Sovereign Wealth Fund*). This model prioritizes flexibility and cost efficiency over physical footprint.

The Chief Executive Officer oversees implementation, while the Board retains ultimate authority over strategy and risk. Crucial to note, but while the government sets the mandate and risk parameters, professional investors retain autonomy over portfolio construction within that framework (*Future Fund Australia Sovereign Wealth Fund*). This separation is foundational to the Fund's ability to operate a Total Portfolio Approach without political interference.

Looking ahead, the Future Fund expects a more challenging investment environment. In response, the Fund is emphasizing flexibility, diversification beyond traditional assets, and granularity in capital allocation. That said, the government's most updated mandate asks the Fund to consider investments in areas such as energy transition, housing, and Australian infrastructure. While this adds additional constraints, the Board has stated clearly that such investments will only be pursued where they meet the Fund's commercial return and risk requirements (*Djurdjevic*).

Overall, the Future Fund's strategy remains consistent with its founding principles: long-term value creation, disciplined risk management, and institutional independence. Its evolution reflects a recognition that modern portfolios must be actively managed, globally diversified, and resilient to structural change, rather than anchored to static allocation models.

Canada Pension Plan Investment Board (CPP Investments)

CPP Investments, formally known as the Canada Pension Plan Investment Board or CPPIB, is Canada's national pension investment manager. It was established in 1997 after demographic projections showed that a pay-as-you-go structure would become unsustainable as Canada's population aged (*CPP Investments*). Prior to reform, CPP contributions were largely invested in provincial bonds at below-market rates, creating long-term solvency risk. The creation of CPPIB separated investment management from government, with a clear mandate to maximize long-term returns. The fund's core objective is to help ensure the long-term sustainability of the CPP for current and future beneficiaries by preserving and enhancing purchasing power over multiple generations, rather than optimizing short-term budget outcomes (*CPP Investments*). The fund's global presence is facilitated by CPPIB's network of offices and partners: it is headquartered in Toronto and maintains offices in major financial centers including New York, London, Hong Kong, Mumbai, Sydney, São Paulo, San Francisco and Luxembourg (*CPP Investments*).

CPP Investments operates with an explicitly long-term horizon and does not have a fixed payout schedule tied to annual government budgets. Contributions and benefits are determined by statute, while CPPIB's role is purely investment management. Clients are the CPP contributors and beneficiaries, represented by governments. Managers of CPP Investments is the CPPIB itself. This separation between the "clients" and the "managers" is central to its governance model. Capital is legally protected from political interference, and assets cannot be withdrawn for non-pension purposes, insulating investment decisions from fiscal cycles and political pressure.

As of March 31, 2025, CPP Investments managed approximately CAD \$714 billion (USD \$517 billion) in assets, making it one of the largest pension funds globally (*CPP Investments*). Assets have grown rapidly from roughly CAD \$36 billion (USD \$24.7 billion) at inception in 1999. While net contributions continue to be positive, the majority of AUM growth over the past decade has been driven by investment returns rather than inflows, reflecting the maturation of the CPP system and the compounding effect of long-horizon investing. For more context, since CPPIB began investing in 1999, investment activities have generated over C\$492 billion in net income, or roughly 70% of total assets (*CPP Investments*).

Performance measurement at CPP Investments is explicitly long-term and total-portfolio focused. The primary objective is to achieve a real inflation-adjusted return of at least 4.0% over the long run, which corresponds to the minimum return required to sustain CPP contribution rates under current actuarial assumptions. Short-term benchmarks are intentionally de-emphasized, as management views annual or even five-year outcomes as noisy and often misleading for a fund with a multi-decade horizon. For the fiscal year ending March 31, 2025, CPP Investments reported a net nominal return of approximately 9.3%. Over the past 10 years, the fund has delivered an annualized net return of roughly 10.1%, and over 15 years approximately 10.5% nominal, translating into strong real returns given Canada's inflation profile over that period (*CPP Investments*). Performance disclosure emphasizes transparency at the total fund level rather than attribution to narrow asset-class benchmarks.

Central to CPP Investments' implementation of a Total Portfolio Approach is its use of a reference portfolio, which acts as a risk anchor rather than a performance benchmark. The current reference portfolio is approximately 85% global equities and 15% nominal bonds, chosen to reflect the fund's long-term risk tolerance and growth orientation(*CPP Investments*). This

reference portfolio defines the level of systematic market risk the Board is willing to accept. All active investment decisions including private equity, infrastructure, real estate, credit, and factor tilts, are evaluated based on whether they improve expected returns or resilience relative to this reference, rather than some predefined asset-class buckets. Capital is allocated dynamically based on relative value, long-term opportunity sets, and the interaction of risks across the portfolio. As of 2025, the fund's actual asset mix includes roughly 27% public equities, 25% private equity, 22% real assets, 19% credit, and the remainder in government bonds and/or cash-like instruments (*CPP Investments*).

Geographically, CPP Investments is highly diversified, with approximately 35% exposure to the Americas, 25% to Europe, and 30% to Asia-Pacific, with the remainder spread across other regions. The United States is the single largest country exposure, reflecting market depth and opportunity density.

CPP Investments' investment philosophy is explicitly long-term, real-return focused, and structurally oriented rather than tactical. The fund does not attempt to time markets in the short run, but it is willing to rebalance aggressively during periods of dislocation, as seen during the Global Financial Crisis and the COVID-19 market shock. Similar to other funds, management has stated that the fund is willing to underperform in late-cycle bull markets if doing so preserves liquidity and downside protection.

On a different note, CPP Investments is structured to support a Total Portfolio Approach. While teams are organized by asset class and geography, compensation and performance evaluation emphasize contribution to total fund outcomes rather than siloed benchmarks. A significant portion of compensation is deferred and tied to long-term results, reinforcing

alignment with the fund's horizon (*CPP Investments*). Governance is handled by an independent professional Board, whose members are appointed through a merit-based process rather than political office. The Board sets risk appetite and oversees management, while the CEO and Chief Investment Officer retain full discretion over investment execution within that framework. This separation between risk-setting and decision-making is a core enabler of CPP Investments' TPA model.

In sum, CPP Investments success rests not on superior asset allocation alone, but on institutional design: governance independence, long-term incentives, internal capability, and a risk framework that treats the portfolio as a single integrated system. While TPA is difficult to replicate, CPP Investments demonstrates that it can deliver durable real returns and stability at scale.

Government of Singapore Investment Corp (GIC)

GIC (Government of Singapore Investment Corporation) is Singapore's sovereign wealth fund. Established in 1981, GIC manages Singapore's foreign reserves with a mandate to preserve and enhance their long-term purchasing power and provide a steady income stream for the government's budget (Menon). Today, it oversees a portfolio estimated at USD \$936 billion in assets as of 2025 (*Ministry of Finance Singapore, 2025b*). GIC's primary performance metric is its 20-year annualized real rate of return, which stood at 3.8% per annum for the period ending March 31, 2025 (*GIC, 2025c*).

GIC maintains a highly diversified portfolio across asset classes and regions. Equities form the largest component of GIC's portfolio, followed by fixed income assets and then real assets. In the latest fiscal year, GIC increased its allocation to equities, while reducing its bond holdings (*GIC, 2025c*). The current asset mix as of March 31, 2025 is approximately as follows:

- Equities: 51% of the portfolio (up from 46% a year earlier)
- Fixed Income: 26% (down from 32% a year earlier)
- Real Assets: 23% (roughly unchanged from 22% the year prior)

This shift reflects GIC's move to capitalize on robust equity markets and a response to rising interest rates that made bonds less attractive. The United States remains GIC's single largest investment market, and overall Americas exposure reached 49% of the portfolio in 2025, which is up from 44% a year earlier (*GIC, 2025c*). In contrast, the fund's allocation to Asia-Pacific declined to about 24%, down from 28%, while Europe, Middle East & Africa held steady at roughly 20% (*GIC, 2025c*). GIC does not disclose country-specific breakdowns, but it has indicated investments in over 40 nations.

As noted before, GIC's 20-year real rate of return is 3.8% per annum for FY2024/25. This is slightly below the 3.9% achieved the previous year, marking the fund's weakest long-term return since 2020, when the 20-year real return hit 2.7% amid the pandemic downturn. As of 2025, the fund's 10-year nominal return was about 5.0% per annum, and the 5-year return was around 6.1% per annum in USD terms (*GIC, 2025c*). These figures indicate that performance in the most recent five-year window has been relatively strong, while the 10-year average is weighed down by weaker early-2010s and pandemic-era returns.

In managing its portfolio, GIC uses a reference portfolio of 65% global equities and 35% global bonds. This serves as a baseline for its risk tolerance and policy goals (Rundell). From this, GIC’s actual portfolio has operated with slightly lower risk resulting in slightly lower returns in comparison to this reference portfolio. As directed by their mandate, GIC has prioritized capital preservation and downside protection to safeguard national reserves (*GIC, 2025c*).

Over the decades, GIC’s portfolio and strategy have evolved considerably. When GIC was first set up in 1981, its mission was to shift Singapore’s reserves from low-yielding, liquid assets into higher-yielding investments for the future (Garten). A major strategic shift came in 2013 with the introduction of a “New Investment Framework” that gave GIC more flexibility to pursue investments that might be riskier in the short term but offer higher long-term returns (*GIC, 2025c*). This marked a deliberate move into areas like private equity, venture capital, real estate, and infrastructure on a larger scale. Over the past decade alone, GIC’s assets under management roughly doubled – it was reported around US\$390 billion in 2018 and is estimated well above US\$800 billion in 2025 (*GIC, 2025c*).

GIC employs approximately 2,000 people globally across 11 offices, making it relatively lean given the scale of assets under management is estimated at over US\$800 billion. The majority of employees work in investment-facing roles, with specialized functions. The GIC’s staffing is distinct in that investment teams are organized by asset class and geography, allowing professionals to develop deep domain expertise rather than rotating frequently across unrelated strategies.

The GIC is headquartered in Singapore, where the majority of senior leadership, governance functions, and central investment strategy teams are based (*GIC, 2025c*). Beyond Singapore, GIC operates a global network of 11 offices in more than 40 markets, enabling proximity to deal flow, partners, and local market intelligence. These office locations include USA, United Kingdom, Germany, France, Japan, China (including Hong Kong), India, and Brazil.

GIC's executive leadership is drawn largely from within the organization, reinforcing institutional continuity and long-term thinking. The Chief Executive Officer oversees day-to-day operations, while the Group Chief Investment Officer coordinates investment activity across asset classes. These executives report to a professional Board that includes senior public officials and private-sector leaders, ensuring alignment with national objectives while preserving operational independence (*Ministry of Finance Singapore, 2025a*). Crucially, investment decisions are not made by politicians. While the Government of Singapore sets overall risk tolerance, GIC's employees—investment professionals, economists, and risk managers—retain autonomy over portfolio construction and asset selection within that framework (Menon). This structure is a core enabler of GIC's ability to execute a Total Portfolio Approach at scale while maintaining discipline, continuity, and political insulation.

Looking ahead, GIC anticipates a more challenging investment environment and is positioning its portfolio accordingly. The fund's managers have warned that returns over the next decade are likely to be more volatile and potentially lower, given today's high asset valuations, higher interest rates, and geopolitical uncertainties in a "world in flux" (*GIC, 2025c*). In response, the fund is emphasizing what it calls "granularity and agility" in its investment strategy (*GIC, 2025a*). Granularity means being very selective and targeted, identifying specific

sub-segments or themes that can outperform. Agility, on the other hand, means maintaining flexibility and liquidity so that GIC can act decisively when market dislocations occur.

This adaptive, forward-looking approach has been a hallmark of GIC's evolution. GIC has consistently balanced its strategy; focusing on long-term value, guarding against permanent capital loss, and proactively structural positioning to ensure that the fund can continue delivering.

CalPERS (California Public Employees' Retirement System, USA)

The California Public Employees' Retirement System, or CalPERS, was established in 1932 to provide retirement security for California public employees. It was later expected to cover public schools and local agencies by 1992, where it was also renamed from State Employees' Retirement System, or SERS, to CalPERS (*CalPERS Annual Comprehensive Financial Report*).

CalPERS core purpose is to invest intentionally to ensure pension benefits can be paid over generations while managing the cost burden on employers and members. That long-duration mandate is visible in the scale of obligations: CalPERS paid \$32.8B in retirement benefits in FY 2023–24 (*CalPERS Annual Comprehensive Financial Report*). Unlike a sovereign wealth fund whose “client” may be a national treasury, CalPERS’ “client” is the membership and participating employers, with the “manager” being the CalPERS Board and Investment Office.

As of FY 2024–25, CalPERS reported a portfolio value of USD \$556.3B. This is a +11.6% net return for the year and surpasses the long-term objective return of 6.8%. Over long

horizons, CalPERS reports an average annual return of about 7.6% over 30 years, crucial for stability and growth to fund liabilities continuously (*CalPERS Annual Comprehensive Financial Report*).

CalPERS discloses results publicly and regularly, reflecting a transparency posture consistent with being a public fiduciary rather than a secrecy-oriented allocator (SAA). The current system has lived through sharp downsides over the years, which is why risk framing and management of such risk has become central to its current evolution.

This risk management reshape guides CalPERS' move toward a Total Portfolio Approach (TPA) on July 1st 2026. In its own Board education materials, CalPERS frames TPA as an evolution from traditional Strategic Asset Allocation (SAA) by shifting the Board's focus from asset-class silos to total portfolio risk, creating a "common purpose" and clearer accountability. The TPA logic is straightforward: the Board sets a risk appetite (through a simple reference portfolio), and management is accountable for active decisions and active risk relative to that reference, rather than being driven by asset-class benchmarks (Gilmore). CalPERS has emphasized that under the old status quo, risk appetite was "not explicitly defined," investments were "driven by asset class benchmarks," and the assumptions underlying fixed SAA could go stale quickly: all issues TPA is designed to correct (Leu et al. 2025).

Under the new model, CalPERS' Board approved a 75% global equity / 25% global bond reference portfolio as the risk anchor, along with a 400 basis points active risk limit. This reference portfolio is meant to make it obvious what portion of results comes from systematic market exposure versus discretionary choices. This is done while ensuring those discretionary choices remain inside a Board-approved risk envelope (Rundell).

In terms of allocation, CalPERS’ published “Facts at a Glance,” which shows a large public equity and fixed income base with meaningful private equity, real assets, and private debt exposures. Geographically, CalPERS is a global investor but remains U.S.-anchored. A concrete, fund-disclosed indicator of “home” exposure is that CalPERS reports about \$79B (14% of assets) invested in California (*CalPERS Annual Comprehensive Financial Report*). The broader implication is that CalPERS must balance global diversification with currency/liquidity needs because benefits are paid in USD and the plan is mature (large ongoing outflows). The geographic stance is therefore best understood as opportunity-driven global investing with a substantial domestic core (Leu et al. 2025).

Organizationally, CalPERS is a large public enterprise with roughly 2,800–2,900 employees overall. Internally, CalPERS has historically managed large portions of public markets in-house, while relying on external partnerships in private markets where specialized sourcing and execution are required (*CalPERS Annual Comprehensive Financial Report*). The critical organizational change implied by TPA is cultural: CalPERS’ describe moving from siloed decision-making to a collaborative “common purpose” and unified language around total-fund risk and return (Gilmore).

Governance is anchored in a 13-member Board of Administration with a mix of elected member representatives, appointed members, and previous state officials. That composition reflects their goal to satisfy public transparency expectations while maintaining fiduciary independence in investment decision-making (*CalPERS Annual Comprehensive Financial Report*). The practical intent of TPA governance is to clarify roles: the Board sets the risk appetite through the reference portfolio and active risk limit, and management is accountable for implementation decisions inside those constraints.

In summary, CalPERS' strategic priorities are to implement TPA to improve resilience in a more volatile regime, increase transparency and accountability, while sustaining return sufficiency toward the 6.8% long-term requirement.

Difficulties with Implementing TPA

Total Portfolio Approach (TPA) is difficult to implement because it is a full restructuring of how a fund is governed, how decisions get made, how risk is defined and monitored, and how the organization evaluates success (Barsoum et al.). In other words, it is not just a better asset allocation philosophy, it's an entirely new operating system. That said, there are several hallmark constraints an organization can face in the attempt to restructure for a Total Portfolio Approach.

The first constraint is governance capacity. SAA is the industry norm because it produces a clean and defensible governance process. The board sets the long-run policy mix, staff implement it, and the outcomes are evaluated against that policy and its related benchmarks (Urwin). Even if SAA is imperfect, it makes oversight straightforward. A trustee can ask, "Are we within our ranges?" "What caused underperformance in public equity?" and "Are we meeting our target return assumption?" Under TPA, those same oversight questions aren't compatible since the framework is explicitly designed to break down the policy buckets and treat the portfolio as one object. That forces a board to move away from familiar controls and instead govern using higher-level concepts like risk budgets, active risk limits, and a reference portfolio. For boards that are not deeply investment-technical, that shift is uncomfortable and harder to execute well. It requires more education, more trust in staff, and stronger institutional discipline, because the board can no longer "audit" the portfolio by checking if each silo did its job.

Even when a fund uses a reference portfolio to anchor governance, oversight can become harder in practice. CalPERS is explicitly trying to manage this by pairing the 75/25 reference with a firm active risk limit so governance remains legible while staff gain flexibility, but that's still a governance leap compared to 11 sleeve benchmarks and detailed policy ranges (CalPERS).

The second challenge is political and regulatory visibility, especially for public funds. SAA functions as a shield: if markets fall, boards and staff can credibly say they followed policy and were diversified in the standard way (Urwin). TPA increases discretion. More discretion means more accountability for the decision itself, not just for the outcome. Any tactical move, deviations from typical investments, or any illiquid allocation can turn into a governance failure instead of a portfolio trade-off. This matters because most public funds live under external scrutiny by the media, legislatures, auditors, beneficiaries, unions, and political cycles. This TPA framework increases the number of defensible, judgment-based decisions that can be “right” economically but still fragile institutionally.

A third barrier is operational infrastructure. TPA requires the fund to have a reliable, decision-useful view of total portfolio risk across public and private assets. Most funds can track public market exposures daily, but private equity, infrastructure, real estate, and private credit are valued infrequently and carry hidden factor exposures that do not show up cleanly in a conventional asset-class report. A fund can't truly run TPA unless it can translate their illiquid exposures into comparable risk terms, and then aggregate those exposures at the total portfolio level (Jarvis). That demands data pipelines, risk systems, scenario analytics, and attribution tooling that many institutions do not have. Furthermore, it also requires staff who can interpret those tools correctly, because risk models are not truth machines. This is part of why full TPA tends to appear in very large funds with serious internal capability and budget.

A fourth problem is organizational structure and incentives. SAA allocates capital and authority, producing an organization where teams own sleeves, have benchmarks, and are rewarded based on relative performance. This structure is stable and career-legible, as an equity head knows what success looks like and the institution knows how to evaluate them (Urwin). TPA challenges this arrangement because it treats all investments as competing for capital based on their marginal contribution to the total fund, not their asset label. This can result in the centralization of more power with the CIO and total-portfolio functions (Barsoum et al.). Even if the fund never acknowledges it explicitly, the implication of allocations can create resistance among teams in private markets where the work is slower-cycle and where the talent market is competitive. If a TPA shift results in reduced or paused commitments for a period, it can trigger retention and morale issues in underperforming teams.

A fifth implementation issue is accountability and attribution. Under SAA, when something goes wrong, it is easy to assign responsibility. Under TPA, the portfolio is intentionally built around interactions, where a strategy can lose money in isolation but be net-positive because it reduces drawdown risk or improves liquidity in stress. While that is the right way to think about a total portfolio, it makes performance explanation harder and can weaken traditional accountability mechanisms. If a board wants to know “who owns this result,” the answer under TPA is often “everyone” (Jarvis).

In short, TPA is difficult to adopt because it requires superior governance, superior modeling systems, and a culture that can handle discretion without losing accountability. SAA persists because it scales across institutions of uneven quality despite TPA only working when the internal institution is built to support it.

Conclusion

This paper shows that the Total Portfolio Approach is not simply a different way to allocate assets, but a different way to run an investment institution. TPA reframes portfolio construction around total-fund objectives, explicit risk appetite, and integrated decision-making, rather than fixed asset-class silos. The experiences of the Future Fund, CPP Investments, and GIC demonstrate that when governance, incentives, and internal capability are aligned, TPA can produce portfolios that are more resilient, more transparent in their risk-taking, and better suited to long-horizon mandates.

At the same time, the analysis makes clear that TPA is difficult to implement and not universally appropriate. Its effectiveness depends on institutional maturity: strong boards, political insulation, robust risk systems, and investment teams capable of operating with discretion. Strategic Asset Allocation persists not because it is optimal in all environments, but because it scales well across institutions with varying governance capacity and provides clear accountability.

CalPERS' planned transition illustrates how TPA is best understood as an evolution rather than a rejection of traditional practice. By introducing a reference portfolio and explicit active risk limits, CalPERS is attempting to preserve governance clarity while gaining the flexibility required to manage total-fund risk more deliberately. Whether this transition succeeds will depend less on portfolio mechanics and more on execution, culture, and discipline over time.

Ultimately, TPA is best viewed as a capability-dependent framework rather than a prescriptive solution. For institutions that can support it, TPA offers a coherent response to changing market structure and longer-dated obligations.

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